Mark Santolucito

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www.marksantolucito.com • October 13, 2026

Research Interests

Program Synthesis, Temporal Logic, Symbolic AI/AGI, Formal Verification, Computer Music

Education

Yale University — New Haven, CT
Computer Science Ph.D. - Advisor: Ruzica Piskac — *May 2020*
PhD Dissertation: A Modular Synthesis Framework for Software Deployment, Design, and Implementation

Yale University — New Haven, CT
Computer Science M.S. - Advisors: Paul Hudak†, Ruzica Piskac — *May 2015*

Amherst College — Amherst, MA
Computer Science B.A. & Music B.A., Cum Laude — *May 2013*
Advisors: Scott Kaplan, Jason Robinson

Professional Experience (Higher Education)

Barnard College, Columbia University — New York, NY
Tenure Track Assistant Professor — *Fall 2020–*

Yonsei University — Seoul, South Korea
Visiting Faculty — *Summer 2023*

Geumgang University — Nonsan, South Korea
Visiting Faculty — *Feb 2016–Aug 2016*

Professional Experience (Industry)

Ndea Inc. — New York, NY
Technical Staff — *March 2025–Sept 2025*
Technical Staff at Ndea, an AGI research lab startup focused on program synthesis.

CaptureAge — New York, NY
Research Engineer — *Dec 2023–Sept 2024*
Consulting on language design for the virtual machine simulator for Age of Empires 2 (1999) replay system.

Amazon — New York, NY
SDE Intern - AWS Security Automation — *Summer 2018*
Applied my research on configuration file analysis to automatically build CloudFormation verification tools for code quality and security.

Academic and Professional Honors

U.S. Fulbright Program
U.S.-Korea Presidential STEM Initiative Award — *Sept 2024–Mar 2025*
Bridging the Semiconductor Culture Gap: Towards Manufacturing-Aware Language Design.

Helmholtz Visiting Researcher Grant
CISPA Center for Information Security — *Jun–Aug 2024*
Provably correct Code Generation with Large Language Models and Reactive Synthesis.

Gladys Brooks Junior Faculty Excellence in Teaching Award
Barnard College — *May 2024*
Recognizes considerable individual achievements of tenure-track/tenure-eligible assistant professors.

Honorable Mention
Google Bughunter Hall of Fame *Nov 2019*
Identified and reported a security vulnerability via Google's Bughunter program on their Infrastructure as Code tool, Cloud Composer.

Program for Excellence in Science
AAAS *Oct 2018*
By nomination of Dean of Graduate Studies, awarded a one-year sponsored membership in AAAS/Science.

Advanced Graduate Leadership Program
Yale University *2018–Present*
$3000 grant to fund additional career development.

Theres and Dennis M. Rohan Fellow
Yale University *2017–2018*
Awarded graduate funding on the recommendation of the Computer Science department.

Heidelberg Laureate Forum
Young Researcher Award *Oct 2017*
Invited with full funding to attend the 5th HLF in Heidelberg, Germany with Turing Award winners and Fields Medalists.

Robert Willets Carle Fellow
Yale University *2014–2015*
Awarded graduate funding on the recommendation of the Computer Science department

Travel Funding Awards
Summer schools: SSFT15, OPLSS2015, SAT/SMT2015, VTSA2017, ProbProg2017
Conferences: USENIX Security 2019, CAV2015/16/17, ICFP2015, POPL2016, FMCAD2016, CHI2019.

Grants

Active Grants

National Science Foundation
$50,000 *2026-2027*
I-Corps: Translation Potential of Verifiable Reward-Signal Generators for Agentic Reinforcement Learning (RL) Environments

National Science Foundation
$590,548 - PI since 2025, transferred from original PI Brian Plancher *2024-2027*
Elements: RUI: Accessible GPU-Accelerated Edge Optimal Control Library and Benchmarks

National Science Foundation
$249,484 *2025-2028*
FMitF: Track III: Introducing Verification Through Creative Embedded Systems

Amazon Web Services
$80,000 *2024-2025*
Statically Inferring Usage Bound for Infrastructure as Code

Prior Awards Now Terminated

Collaboratory Fund (Columbia Data Science Institute & Columbia Entrepreneurship)
$104,019 - with CoPI Seth Cluett *2021-2024*
Accessible and Inclusive Data Capture and Display: Creative Embedded Systems for Multi-Sensory Data Engagement

National Science Foundation
$173,341 *2021-2024*
CRII: SHF: RUI: Exploring Human-in-the-loop Program Synthesis Through Live Coding

Fund for Innovation in Teaching
Barnard College Provost's Office *Fall 2021*
$6,000 - *Arts and Computing in NYC*, for a cross-institutional course with the Fashion Institute of Technology

Student Travel Grant (CCF-2122164)
National Science Foundation *2021-2022*
$10,000 - Student Travel Grant for 2021 Formal Methods in Computer-Aided Design (FMCAD)

Under Submission / In Preparation

Warm starting state-space models with automata learning.
William Fishell, Sam Nicholas Kouteili, and *Mark Santolucito*.
arXiv, 2026.
under submission.

TempoBench: Reasoning execution without causal attribution is just simulation.
Nikolaus Holzer, William Fishell, Baishakhi Ray, and *Mark Santolucito*.
arXiv, 2026.
under submission.

Program synthesis.
Elizabeth Polgreen, *Mark Santolucito*, and Armando Solar-Lezama.
In *Handbook of Model Checking*. Springer, 2nd edition, 2026.
in preparation, invited book chapter.

Publications ($\downarrow_Z^A$ indicates alphabetic author ordering, otherwise senior/primary advising author is last)

Mining beyond the bools: Learning data transformations and temporal specifications.
Sam Nicholas Kouteili, William Fishell, Christian Scaff, *Mark Santolucito*, and Ruzica Piskac.
In *European Symposium on Programming (ESOP@ETAPS)*, 2027.
To Appear.

Lattice deduction transformers.
Liam Davis, Leopold Haller, Alberto Alfarano, and *Mark Santolucito*.
In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
To Appear.

PALS: Preference-guided active automata learning for symbolic reinforcement learning in games.
William Fishell, Sam Nicholas Kouteili, *Mark Santolucito*, and Christian Scaff.
In *Workshop on New Frontiers in Game-Theoretic Learning (NExT-Game@ICML)*, 2026.

Measuring and reducing semantic entropy in LTL guardrail generation.
Maria Hwang, Rachel Papirmeister, and *Mark Santolucito*.
In *International Conference on Verified Software: Theories, Tools, and Experiments (VSTTE)*, 2026.
To Appear.

Learning GR(1) specifications from traces.
Sam Nicholas Kouteili, William Fishell, *Mark Santolucito*, and Ruzica Piskac.
In *Formal Methods in Computer Aided Design (FMCAD)*, 2026.

User engagement with intermediate representations in LLM-assisted audio programming.
Chris Hyorok Lee, Miles Scharff, Katie Wang, and *Mark Santolucito*.
In *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*, 2026.
Short paper. Also presented at the Workshop on Generative AI, Creativity, and Human-AI Co-Creation at ICML 2026.

Blindmarket: Enabling verifiable, confidential, and traceable ip core distribution in zero-trust settings.
Zhaoxiang Liu, Samuel Judson, Raj Dutta, *Mark Santolucito*, Xiaolong Guo, and Ning Luo.
In *Hardware-Oriented Security and Trust (HOST)*, 2026.
To Appear.

Experiential learning of runtime monitoring using pachinko.

Miles Scharff, Maria Chemodanova, and *Mark Santolucito*.
In *Tools for Educational Activities in Logic (TEAL@FLoC)*, 2026.

Automated power domain insertion and control in dataflow circuits.
Martha Barker, *Mark Santolucito*, Stephen A. Edwards, and Martha A. Kim.
In *MEMOCODE*. ACM, 2025.

Embedding alignment in code generation for audio.
Sam Kouteili, Hiren Madhu, George Typaldos, and *Mark Santolucito*.
CoRR, abs/2508.05473, 2025.
Also presented at AI for Music workshop at NEURIPS 2025.

Statically inferring usage bounds for infrastructure as code.
Feitong Qiao, Aryana Mohammadi, Jurgen Cito, and *Mark Santolucito*.
In *International Conference on Verified Software: Theories, Tools, and Experiments (VSTTE)*, Oct 2024.

Towards Reactive Synthesis as a Programming Paradigm.
Angel Leyi Cui, Raven Rothkopf, and *Mark Santolucito*.
In *PLATEAU Workshop 2024*, March 2024.

Poster: Blindmarket: A trustworthy chip designs marketplace for ip vendors and users.
Zhaoxiang Liu, Ning Luo, Samuel Judson, Raj Gautam Dutta, Xiaolong Guo, and *Mark Santolucito*.
In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2024.

Maxpy: An open-source python package for programmatic construction and manipulation of maxmsp patches.
Ranger Liu, Satch Peteron, Richard Lee, and *Mark Santolucito*.
In *New Interfaces for Musical Expression NIME 2023*, October 2023.

Towards the Usability of Reactive Synthesis: Building Blocks of Temporal Logic.
Raven Rothkopf, Angel Leyi Cui, Hannah Tongxin Zeng, Arya Sinha, and *Mark Santolucito*.
In *PLATEAU Workshop 2023*, March 2023.

Can reactive synthesis and syntax-guided synthesis be friends?.
Wonhyuk Choi, Bernd Finkbeiner, Ruzica Piskac, and *Mark Santolucito*.
In *Programming Language Design and Implementation PLDI*, March 2022.

Learning CI configuration correctness for early build feedback.
Mark Santolucito, Jialu Zhang, Ennan Zhai, Jürgen Cito, and Ruzica Piskac.
In *IEEE International Conference on Software Analysis, Evolution and Reengineering SANER*, March 2022.

Program synthesis for musicians: A usability testbed for temporal logic specifications.
Wonhyuk Choi, Michel Vazirani, and *Mark Santolucito*.
In *Programming Languages and Systems - 19th Asian Symposium, APLAS*, October 2021.

Demo: Synthesis-enabled live coding on the web.
Kat Pompermayer, Catherine Ji, Hannah Macias, and *Mark Santolucito*.
In *Programming Languages and Interactive Entertainment PLIE 2021*, October 2021.

Human-in-the-loop program synthesis for live coding.
Mark Santolucito.
In *FARM 2021: Proceedings of the 9th ACM SIGPLAN International Workshop on Functional Art, Music, Modelling, and Design*, August 2021.

The FMCAD 2021 student forum.
Mark Santolucito.
In *Formal Methods in Computer Aided Design, FMCAD*, July 2021.

cardcomposer: A functional programming card game.
Maria Hwang and *Mark Santolucito*.
In *ITiCSE 2021: 26th ACM Conference on Innovation and Technology in Computer Science Education*, June 2021.

TSL synthesis synthesizer: Reconfigurable signal flows through program synthesis.
Michel Vazirani, Wonhyuk Choi, and *Mark Santolucito*.
In *New Interfaces for Musical Expression NIME*, June 2021.

Analyzing infrastructure as code to prevent intra-update sniping vulnerabilities.
Julien Lepiller, Ruzica Piskac, Martin Schäf, and *Mark Santolucito*.
In *Tools and Algorithms for the Construction and Analysis of Systems (TACAS@ETAPS)*, 2021.

Live coding sequencers.
Mark Santolucito.
Hybrid Live Coding Interfaces: performance and craft, July 2020.
https://hybrid-livecode.pubpub.org/pub/live-coding-sequencers.

Formal methods and computing identity-based mentorship for early stage researchers.
Mark Santolucito and Ruzica Piskac.
In *ACM Technical Symposium on Computer Science Education (SIGCSE)*, March 2020.

Grammar filtering for syntax-guided synthesis.
Kairo Morton, Bill Hallahan, Elven Shum, Ruzica Piskac, and *Mark Santolucito*.
In *The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI)*, February 2020.

Software engineering for infrastructure and configuration (SEConfig) - workshop report.
Jürgen Cito and *Mark Santolucito*.
ACM SIGSOFT Softw. Eng. Notes, 2020.

Towards checkpoint placement for dynamic memory allocation in intermittent computing.
Nicholas Shoemaker, Ruzica Piskac, and *Mark Santolucito*.
In *Tools for Automatic Program Analysis (TAPAS@SPLASH)*, 2020.

Synthesizing functional reactive programs.
Bernd Finkbeiner, Felix Klein, Ruzica Piskac, and *Mark Santolucito*.
In *Haskell Symposium*, October 2019.

System design with TSL.
Bernd Finkbeiner, Felix Klein, Ruzica Piskac, and *Mark Santolucito*.
In *SYNT workshop at CAV*, July 2019.

Temporal stream logic: Synthesis beyond the bools.
Bernd Finkbeiner, Felix Klein, Ruzica Piskac, and *Mark Santolucito*.
In *International Conference on Computer Aided Verification (CAV)*, July 2019.

Live programming by example.
Mark Santolucito, William T. Hallahan, and Ruzica Piskac.
In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*, May 2019.

Studio report: Yale open music initiative.
Scott Petersen, *Mark Santolucito*, and Konrad Kaczmarek.
In *International Computer Music Conference (ICMC)*, 2019.

Programming by example: Efficient, but not "helpful".
Mark Santolucito, Drew Goldman, Allyson Weseley, and Ruzica Piskac.
In *PLATEAU at SPLASH*, 2018.
Also presented at SYNT 2018.

Programming-by-example for audio: Synthesizing digital signal processing programs.
Mark Santolucito, Kate Rogers, Aedan Lombardo, and Ruzica Piskac.
In *Functional Art and Music (FARM) at ICFP*, 2018.

Synthesizing configuration file specifications with association rule learning.
Mark Santolucito, Ennan Zhai, Rahul Dhodapkar, Aaron Shim, and Ruzica Piskac.
Proc. ACM Program. Lang., 1(OOPSLA), October 2017.

↓A Z Vehicle platooning simulations with functional reactive programming.
Bernd Finkbeiner, Felix Klein, Ruzica Piskac, and *Mark Santolucito*.
In *Safe Control of Autonomous Vehicles Workshop at CPSWeek*, 2017.
https://arxiv.org/abs/1803.10383.

Version space learning for verification on temporal differentials.
Mark Santolucito.
In *International Symposium on Software Testing and Analysis (ISSTA)*, 2017.
Also presented as poster at FMCAD 2016 Student Research Competition, 3rd Place Award.

Probabilistic automated language learning for configuration files.
Mark Santolucito, Ennan Zhai, and Ruzica Piskac.
In *International Conference on Computer Aided Verification (CAV)*, 2016.

↓A Z Real-time interactive music in haskell.
Paul Hudak, Donya Quick, *Mark Santolucito*, and Daniel Winograd-Cort.
In *Functional Art and Music (FARM) at ICFP*, 2015.

Using javascript as an intermediate language for FRP.
Mark Santolucito and Ruzica Piskac.
Poster at ICFP Student Research Competition, 2015.

Media Modules: Intermedia Systems in a Pure Functional Paradigm.
Mark Santolucito, Donya Quick, and Paul Hudak.
In *International Computer Music Conference (ICMC)*, 2015.

Raid the fridge!: Promoting healthy eating habits through the game Monster Appetitie.
Maria Hwang, Pantiphar Chantes, and *Mark Santolucito*.
Extended Abstract and Poster at Games Learning and Society 10, Best in Show Award, 2014.

Communalizing the interfaces of single player games.
Mark Santolucito and Maria Hwang.
Extended abstract in Digital Games Research Association Conference, 2014.

Simquabbin project: Game-based environmental science education in a virtual world.
Mark Santolucito and Scott Payne.
Exteneded Abstract and Poster at Games Learning and Society 9, 2013.

Designing a community to support long-term interest in programming for middle school children.
Kyle J. Harms, Jordana H. Kerr, Michelle Ichinco, *Mark Santolucito*, Alexis Chuck, Terian Koscik, Mary Chou, and Caitlin L. Kelleher.
In *Proceedings of the 11th International Conference on Interaction Design and Children*, IDC '12, 2012.

Talks

Invited

Dagstuhl Seminar 26421: Advancing Testability and Verifiability of CPS with Neurosymbolic and Large Language Models — *Oct 2026*
50 Years after the Church Synthesis Problem

Ndea *Mar 2025*
An Academic History of Program Synthesis

Fulbright Commission in Seoul *Nov 2024*
Bridging the Semiconductor Culture Gap

Yonsei University *Nov 2024*
Reactive Synthesis as a Programming Paradigm

Dagstuhl Seminar 24171: Automated Synthesis: Functional, Reactive and Beyond *Apr 2024*
Reactive Program Synthesis Modulo LLM Code Generation

Dagstuhl Seminar 23391: The Futures of Reactive Synthesis *Oct 2023*
Reactive Synthesis as a Programming Paradigm

Dagstuhl Seminar 23082: Resilient Software Configuration and Infrastructure Code Analysis *Feb 2023*
Static Analsis for Infrastructure as Code

Isaac Newton Institute Seminar Series: Synthesis United *July 2022*
Can reactive synthesis and syntax-guided synthesis be friends?

Harness.io *Jan 2020*
Program Synthesis for Software Systems

Facebook Faculty Networking & Communications Event, San Francisco, CA *June 2019*
Automated Firewall Repair and Verification.

Xerox PARC, Palo Alto, CA *Oct 2018*
Language Learning for Verification of Configuration Files

Verification and Synthesis for Software Evolution at ETAPS, Greece *Apr 2018*
Learning Models of Configuration Correctness.

Learning in Verification Workshop at ETAPS, Greece *Apr 2018*
Using Machine Learning to Synthesize Specifications for Configuration Files

Instituto Superior Técnico (IST), Portugal *May 2017*
Language Learning for Verification of Configuration Files.

Saarland University, Germany *Sept 2016*
Verifying Configuration Files with Examples.

Monthly Music Hackathon, NYC, NY *Jan 2015*
Workshop on Algorithmic Composition with Euterpea.

Symposium

Shonan (No.235) *Mar 2026*
Automata Methods for Neurosymbolic AI

IFIP WG 2.3 (Observer) *May 2025*
Using Program Synthesis for ARC-AGI

IFIP WG 2.3 (Observer) *May 2024*
Reactive Synthesis as a Programming Paradigm

IBM PL Day, Yorktown Heights, NY *Dec 2018*
Learning to Verify Infrastructure as Code.

New England Programming Languages Symposium, Cambridge, MA *Sept 2018*
Digital Signal Processing Programming-by-Example.

IBM PL Day, Yorktown Heights, MA *Dec 2017*

Synthesizing Functional Reactive Programs.

Teaching and Supervising Experience

Courses designed and taught

MUSI GR6610: Live Coding: Code as Performance — *Columbia University*
Fall 2026

COMS BC3997: New Directions in Computing: Neurosymbolic AI Seminar — *Barnard College*
Fall 2026

COMS BC3930: Creative Embedded Systems — *Barnard College*
Spring 2021, 2022, 2023, 2024, 2026

COMS BC3430: Computational Sound — *Barnard College*
Fall 2020, Fall 2021, Fall 2022, Fall 2023, Spring 2024, Spring 2026

COMS W1002: Computing in Context — *Columbia University*
Fall 2025

COMS W1002: Computing in Context: Computing in the Arts Track — *Columbia University*
(Track Instructor) Fall 2020, 2021, 2022, 2023, 2025, 2026

IEE1164: Computing in Context: Computing in the Arts — *Yonsei University*
Summer 2023

COMS BC3997: New Directions in Computing: Program Synthesis Seminar — *Barnard College*
Spring 2023

COMS BC3997: New Directions in Computing: Arts and Computing in NYC — *Barnard College*
cross-institutional with Fashion Institute of Technology. Fall 2021

COMS BC3997: New Directions in Computing: Audio Plugin Development — *Barnard College*
Spring 2021

COMS BC1002: Methods and Problems in Computer Science — *Barnard College*
Summer 2021

CPSC334: Creative Embedded Systems — *Yale University*
(Co-Instructor) Fall 2019

CS101: Intro to Computer Science — *Geumgang University*
Spring 2016

CS201: Object Oriented Programming — *Geumgang University*
Spring 2016

CS032: Computer Music — *Geumgang University*
Spring 2016

Reading groups
Category Theory for Computer Scientists - reading from Riehl and Awodey (Summer and Fall 2021), with Columbia graduate students and Barnard undergraduates.

PhD Thesis Committees
Andoni Rodriguez, IMDEA Software Institute, Madrid, Spain (advised by Cezar Sanchez), 2026 (proj.)
Maxwell Levatich, Columbia University, USA (advised by Stephen Edwards), 2026 (proj.)
Vikram Nitin, Columbia University, USA (advised by Baishakhi Ray), 2026
Martha Baker, Columbia University, USA (advised by Matha Kim), 2025

MS Thesis Committees

Feitong Qiao, Columbia University, USA, 2024
Julia Laichner, CIPSA, Saarland University, Germany, 2025

Student co-author mentees
Feitong Qiao (Columbia MS) [14]: Summer 2022 - Summer 2024
Angel Cui (Barnard/Columbia MS → CMU PhD) [18, 15]: Summer 2022 -
Raven Rothkopf (Barnard → UCSD PhD, GRFP Fellow) [18, 15]: Summer 2022 - Summer 2024
Hannah Zeng (Barnard) [18]: Summer 2022 - Summer 2023
Richard Lee (Columbia) [17]: Summer 2022 - Fall 2022
Satch Peterson (Columbia) [17]: Summer 2022 - Fall 2022
Ranger Liu (Columbia → GRFP Fellow) [17]: Summer 2022 - Fall 2022
Wonhyuk Choi (Columbia MS → Meta) [21, 26]: Fall 2020 - Fall 2021
Michel Vazirani (Columbia) [21, 26]: Fall 2020 - Summer 2021
Kat Pompermayer (Barnard) [22]: Summer 2021
Catherine Ji (Barnard) [22]: Summer 2021
Hannah Macias (Barnard) [22]: Summer 2021
Nicholas Shoemaker (Yale) [32]: 2019
Kairo Morton (MIT) [30]: 2019
Elven Shum (Yale) [30]: 2019
Drew Goldman (Roslyn High School) [38]: 2018
Kate Rogers (Yale) [39]: 2017
Aeden Lombardo (Yale) [39]: 2017
Aaron Shim (Yale) [40]: 2016

Service

University Service

Barnard College
Science Pathways Scholars Program (SP2). Computer Science Representative. (2021-)
IT Committee. (2022-2025).
Presidential Artificial Intelligence and Technology Task Force. (2024).
Faculty Governance and Procedures (FGP) Committee. (2021-2023).
Co-founder with Lydia Chilton and faculty advisor for Design at Columbia. (2020-2023)
Columbia University
Computer Science DEI Coordination Committee. Barnard Representative. (2022-2024)
Computer Science Academic Committee. Barnard (non-voting) Representative. (2022-2024)

Professional Service

Served on NSF Panel in 2021

Organization
1. The 27th Conference on Formal Methods in Computer Aided Design (FMCAD 2027), hosted at Barnard College, New York, NY, October 2027. Co-chair
2. Synthesis Workshop 2025, co-organizer with Shufang Zhu (SYNT 2025)
3. TACAS Artifact Evaluation, co-chair with Daniela Kaufmann (TACAS 2025)
4. Dagstuhl Seminar 23082: Resilient Software Configuration and Infrastructure Code Analysis, co-organizer with Jürgen Cito, Ruzica Piskac, Andy Zaidman, and Yuanyuan Zhou
5. The 23rd International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI 2022). Artifact Evaluation Chair
6. The 21st Conference on Formal Methods in Computer Aided Design (FMCAD 2021). Student Forum Chair

7. CONFLANG 2021, co-organizer with Jürgen Cito, Eelce Dolstra, Gabriella Gonzalez, Yann Hamdaouim, Nicolas Jeannerod, Marcel van Lohuizen
8. SEConfig 2019, co-organizer with Jürgen Cito
9. The 5th ACM SIGPLAN International Workshop on Functional Art, Music, Modelling and Design (FARM 2017). Publicity Chair
10. The 4th ACM SIGPLAN International Workshop on Functional Art, Music, Modelling and Design (FARM 2016). Publicity Chair
11. CAV Buddy System 2015, 2016, 2017

Program Committees

12. AAAI Conference on Artificial Intelligence (AAAI 2026)
13. International Conference on Verified Software: Theories, Tools, and Experiments (VSTTE 2026)
14. Object-oriented Programming, Systems, Languages, and Applications (OOPSLA 2026)
15. Conference on Formal Methods in Computer Aided Design (FMCAD 2025, 2026)
16. 37th International Conference on Computer Aided Verification (CAV 2025)
17. Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2024, 2025)
18. International Conference on Networking, Architecture, and Storage (NAS 2024)
19. Object-oriented Programming, Systems, Languages, and Applications (OOPSLA 2023)
20. Workshop on Programming Languages and Interactive Entertainment (PLIE 2021)
21. Workshop on Reactive and Event-based Languages & Systems (REBLS 2020, 2021)
22. The 8th Workshop on Synthesis (SYNT 2019)
23. The 2nd International Workshop on Machine Learning techniques for Programming Languages (ML4PL 2018)

Journal Referee

24. Formal Methods in System Design (FMSD 2026)
25. Journal of Automated Reasoning (JAR 2021, 2026)
26. Transactions on Software Engineering (TSE 2024, 2025)
27. Acta Informatica (2021)
28. IEEE Signal Processing Letters (IEEE SPL 2021)
29. ACM Transactions on Programming Languages and System (TOPLAS 2017)

Reviewer

30. Technical Symposium on Computer Science Education (SIGCSE 2020, 2021, 2022).
31. New Interfaces for Musical Expression (NIME 2019, 2020, 2021)
32. The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT 2019).

Subreviewer

iFM 2018, SMT 2017, ESOP 2017, ICDCIT 2016, VSTTE 2015

Artifact Evaluation Committee

USENIX Security 2020 Fall, PLDI 2018

Community Outreach Talks and Workshops

Abstract Synthesis (Podcast) *Sept 2025*
Grammer Filtering for Syntax Guided Synthesis

Computer Music Center, Columbia University, NYC, NY *Mar 2022*
Guest lecture in MUSI GR6602, Sonic and Visual Representation of Data

Barnard Computational Science Center, NYC, NY *Dec 2021*
Live Coding: Play Your Laptop Like an Instrument

Teachers College, Columbia University, NYC, NY *Nov 2021*

Guest lecture in MSTU 4133, Computing and Cognition

Fashion Institute of Technology, NYC, NY *Oct 2019*
Programming Digital Embroidery

NYC CS Fair, NYC, NY *Mar 2019*
How to Play Your Laptop Like an Instrument: Live Coding for Music

Yale Computer Science Society, New Haven, CT *Feb 2018*
Panel Discussion - An Inside Look: CS Graduate School

Code441 Hackathon, Hamilton, Bermuda *Dec 2018*
Applications of Association Rule Learning and Neural Networks

Roslyn High School, NY *Nov 2017*
Majoring in Computer Science - the Why and How.